

AMOLDIP SINGH NARANG

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Summary

Experienced in semiconductor device physics, cleanroom operations, and process optimization. Skilled in data analysis, thin film deposition, and material characterization. Proficient in simulation tools, driving efficiency and yield improvement. Collaborative and results-driven, delivering impactful solutions in research and industrial environments.

Skills

- Semiconductor Device Physics
- Process Optimization
- Cleanroom Operations
- Data Analysis (JMP, FabGuard, SPC)
- Catalyst Synthesis
- Material Characterization (SEM, XRD, AFM)
- Simulation Software (ANSYS, SolidWorks, MATLAB)
- Thin Film Deposition (PECVD, MBE, Sputtering)
- HVAC Systems Design
- Corrosion Prevention Solutions
- Photolithography Optimization
- Green Technology Implementation

Experience

Visiting Research Scholar (Affiliate Position) Tufts University, Medford, MA	10/2024 to Current
- Enhanced understanding of two-photon polymerization for rapid prototyping of micro-needles on silicon substrates, improving biomedical device design. - Applied nanoscribe technology to fabricate optical devices, achieving sub-micron resolution and reducing development time by 30%. - Analyzed feature size precision using advanced imaging techniques, improving measurement accuracy by 15%. - Optimized fabrication parameters, increasing throughput by 20% in prototype development.	
Process Engineer (Co-Op) Skyworks Solutions, Woburn, MA	05/2023 to 12/2023
- Developed wafer fabrication processes for RF devices, achieving a 15% reduction in defect rates through collaboration with senior engineers. - Improved TaN thin film uniformity by 25% via PECVD process optimization, increasing device yield. - Conducted statistical analysis correlating tool performance with wafer metrics, enhancing production line efficiency by 20%. - Designed DOE experiments optimizing photolithography and etching processes, reducing cycle time by 10%.	
Research Assistant Tufts University, Medford, MA	08/2023 to 01/2024
- Investigated pore confinement impact on catalyst selectivity, boosting propane-to-propene conversion efficiency by 25% using gas chromatography. - Synthesized catalysts and analyzed data to enhance reactivity metrics for acrolein and acrylic acid production. - Collaborated with interdisciplinary teams to apply catalytic research insights to semiconductor manufacturing challenges. - Published findings on catalyst-material interaction, contributing to lab's research impact metrics.	
Industrial Training Intern Academy of Skill Development	05/2021 to 07/2021 Noida, India
- Conducted CFD simulations for multiphase flows, streamlining design processes for industrial fluid systems. - Delivered a project on ANSYS and SolidWorks simulations, enhancing design accuracy by 20%. - Collaborated on multidisciplinary research agendas, promoting advanced modeling tools for practical industrial applications. - Presented results in technical workshops, improving training module adoption rates by 15%	
Intern Bharat Petroleum Corporation Ltd	07/2019 to 08/2019 Noida, India
- Designed HVAC systems integrating Variable Frequency Drives, reducing annual energy costs by 20%. - Researched ethanol production technologies, optimizing second-generation processes for sustainable fuel development. - Evaluated corrosion-resistant coatings, improving crude oil pipeline durability by 30%. - Generated technical reports for management, expediting R&D project approvals by 10%	

Intern**Indian Oil Corporation Ltd****01/2017 to 02/2017****Delhi, India**

- Analyzed refining process efficiency, identifying 10% optimization potential in by-product separation.
- Conducted green technology feasibility studies, contributing to industry-wide sustainability benchmarks.
- Recommended process improvements for upstream and downstream operations, reducing waste generation by 15%.
- Presented findings to senior executives, influencing strategic decisions in refinery modernization.

EDUCATION

Tufts University, Medford, MA*Master of Science in Materials Science and Engineering – May 2024*

Relevant Courses: VLSI, Nanophotonics, Micro/Nanofabrication, Semiconductor Device Lab, Optoelectronic Characterization

National Institute of Technology, Srinagar, India*Bachelor of Technology in Chemical Engineering – June 2021*

Relevant Courses: Process Fluid Mechanics, Thermodynamics, Plant Economics, Heat Transfer

PROJECTS

- Simulated thermo-photovoltaic solar cells in TCAD, comparing InGaAs and GaSb efficiency across layer configurations.
- Electrically characterized atomic-scale structures using AFM, optimizing material properties for semiconductor applications.
- Investigated graphene quantum dots for optoelectronic devices, enhancing device performance by 15%.
- Modeled non-Newtonian fluid flows in ANSYS, refining crude oil transportation methods for cost and efficiency gains.

ADDITIONAL INFORMATION

- Open to OPT training opportunities, no immediate sponsorship needed.
- Available to begin employment immediately.